

Correctness Proof of the Bellman–Ford Shortest Path Algorithm

Setup and Definitions

Let $G = (V, E, w)$ be a weighted directed graph where $w : E \rightarrow \mathbb{R}$ assigns arbitrary real-valued weights to edges (negative weights are permitted). Fix a source vertex $s \in V$. Let $n = |V|$.

Definition 1. The *shortest-path distance* from s to v is

$$\delta(s, v) = \begin{cases} \min\{w(p) : p \text{ is a path from } s \text{ to } v\} & \text{if such a path exists} \\ \text{and no negative cycle is reachable from } s \text{ to } v, & \\ -\infty & \text{if a negative cycle is reachable on } s \text{ to } v, \\ \infty & \text{if no path from } s \text{ to } v \text{ exists.} \end{cases}$$

Definition 2. A *negative cycle* is a cycle whose total edge weight is strictly negative.

Algorithm

Algorithm 1 Bellman-Ford(G, s)

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1:  $d[s] \leftarrow 0; \quad d[v] \leftarrow \infty$  for all  $v \neq s$ 
2: for  $i = 1$  to  $n - 1$  do ▷  $n - 1$  relaxation passes
3:   for each edge  $(u, v) \in E$  do
4:     if  $d[u] + w(u, v) < d[v]$  then
5:        $d[v] \leftarrow d[u] + w(u, v)$  ▷ relax edge  $(u, v)$ 
6:     end if
7:   end for
8: end for
9: for each edge  $(u, v) \in E$  do ▷ negative-cycle detection
10:  if  $d[u] + w(u, v) < d[v]$  then
11:    report “negative cycle reachable from  $s$ ”
12:  end if
13: end for

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The outer loop runs exactly $n - 1$ times, relaxing every edge in each pass.

Correctness

We prove that if no negative cycle is reachable from s , then after $n - 1$ passes $d[v] = \delta(s, v)$ for every vertex v reachable from s .

Theorem 1 (Correctness — No Negative Cycle). *Assume no negative cycle is reachable from s . After $n - 1$ passes of the outer loop, $d[v] = \delta(s, v)$ for every vertex v reachable from s .*

Proof. We use strong induction on i to prove the following invariant:

Claim. After i iterations, for every vertex v reachable from s by a shortest path using at most i edges,

$$d[v] = \delta(s, v).$$

Base case ($i = 0$). The only vertex reachable from s with 0 edges is s itself. We have $d[s] = 0 = \delta(s, s)$. ✓

Inductive hypothesis. Assume that after i iterations, for all vertices v reachable from s by a path of at most i edges,

$$d[v] = \delta(s, v).$$

Inductive step. Let v be any vertex reachable from s via a shortest path P using exactly $i + 1$ edges:

$$P = \langle s, \dots, u, v \rangle.$$

The sub-path from s to u is itself a shortest path to u (by the *optimal substructure* of shortest paths) and uses at most i edges. By the inductive hypothesis, after i iterations:

$$d[u] = \delta(s, u).$$

In the $(i + 1)$ -th iteration every edge is relaxed, so in particular edge (u, v) is relaxed. Since $d[v] \geq \delta(s, v)$ always holds (relaxations only assign weights of explicit paths, never underestimating), the relaxation gives:

$$\begin{aligned} d[v] &\leq d[u] + w(u, v) \\ &\leq \delta(s, u) + w(u, v) \\ &= \delta(s, v). \end{aligned}$$

Combined with $d[v] \geq \delta(s, v)$, we conclude $d[v] = \delta(s, v)$ after $i + 1$ iterations. The invariant is maintained.

Conclusion. Since no negative cycle is reachable from s , every shortest path is a simple path. A simple path on $n = |V|$ vertices uses at most $n - 1$ edges. Applying the claim with $i = n - 1$:

$$d[v] = \delta(s, v) \quad \text{for all reachable } v.$$

□

Negative-Cycle Detection

Theorem 2 (Negative-Cycle Detection). *After $n - 1$ passes, there exists an edge $(u, v) \in E$ with $d[u] + w(u, v) < d[v]$ if and only if a negative cycle is reachable from s .*

Proof. ($\Leftarrow$) Suppose no negative cycle is reachable from s . By Theorem 1, $d[v] = \delta(s, v)$ for all v . For any edge (u, v) , the triangle inequality for shortest paths gives $\delta(s, v) \leq \delta(s, u) + w(u, v)$, so $d[v] \leq d[u] + w(u, v)$, meaning no edge triggers the detection condition.

($\Rightarrow$) Suppose edge (u, v) satisfies $d[u] + w(u, v) < d[v]$ after $n - 1$ passes. Since every $d[\cdot]$ value is the weight of some explicit path from s , we always have $d[v] \geq \delta(s, v)$. Because $d[v]$ would still be decreased by a further relaxation, the distances have not stabilised, implying that shortest paths require more than $n - 1$ edges. Any walk of length $\geq n$ on n vertices must repeat a vertex, forming a cycle. If that cycle had non-negative weight, removing it would yield a shorter or equal-length walk, contradicting the assumption that $n - 1$ passes were insufficient. Hence the cycle has strictly negative weight and is reachable from s (it lies on a path that s can reach). $\square$

Corollary 1. *If the detection step reports a negative cycle, then $\delta(s, v) = -\infty$ for some $v \in V$; otherwise the values $d[v]$ returned are exact shortest-path distances.*

Why Negative Weights Are Handled Correctly

Unlike Dijkstra's algorithm, Bellman–Ford does not assume non-negative weights. The key structural difference is that Bellman–Ford relaxes *every* edge in each pass, allowing earlier overestimates to be corrected retroactively in later passes. The induction above tracks this process: after i passes, any vertex whose shortest path uses $\leq i$ edges has its distance computed exactly, regardless of the sign of the weights along it.

Complexity

The outer loop executes $n - 1$ times; each pass iterates over all $|E|$ edges. The detection step is an additional single pass over $|E|$ edges.

$$T(|V|, |E|) = O(|V| \cdot |E|).$$

This is worse than Dijkstra's $O((|V| + |E|) \log |V|)$ on non-negative graphs, but Bellman–Ford handles negative weights and detects negative cycles, which Dijkstra cannot.